serviceOrchestrationLockManagement

Service Description

Service ID: *"serviceOrchestrationLockManagement"*

Abstract
This document describes the `serviceOrchestrationLockManagement` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceOrchestrationLockManagement` service, which provides operator management of orchestration locks, the exclusive reservations of service instances. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

New in Arrowhead 5.

2.2 How This Service Is Meant to Be Used

When a consumer requests exclusivity (`exclusivityDuration`), DynamicServiceOrchestration locks the chosen instance for it. Operators can also create locks, query them and remove them.

2.3 Important Delimitations and Dependencies

Only providers whose service instance metadata allows exclusivity can be reserved. Locks expire and are removed by the cleaner job.

3 Service Interface

This section describes the interfaces to the `serviceOrchestrationLockManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

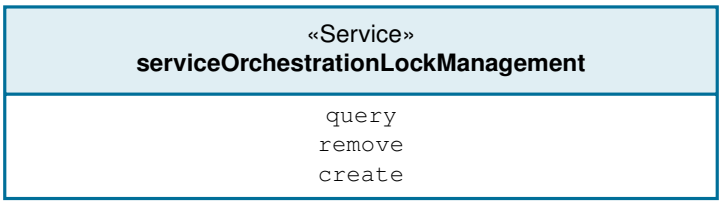


Figure 1: SysML-style block of the `serviceOrchestrationLockManagement` service and its operations.

The following interface operations are available.

3.1 operation query (OrchestrationLockQueryRequest) : OrchestrationLockListResponse

Returns a page of locks matching the filter.

3.2 operation remove (Name)

Removes the locks of an owner on the given service instances.

3.3 operation create (OrchestrationLockListRequest) : OrchestrationLockListResponse

Creates locks on service instances for owners.

4 Information Model

Here, all data objects that can be part of the `serviceOrchestrationLockManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.7, which are used to represent things like hashes and identifiers.

4.1 struct OrchestrationLockListRequest

Locks to create.

Field	Type	Description
locks	List<OrchestrationLockRequest>	Locks to create.

4.2 struct OrchestrationLockListResponse

List of orchestration locks.

Field	Type	Description
entries	List<OrchestrationLockResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.3 struct OrchestrationLockQueryRequest

Paged filter for orchestration locks.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
ids	List<Number>	Lock identifiers to match.
orchestrationJobIds	List<Identifier>	Orchestration job identifiers to match.
serviceInstanceIds	List<Identifier>	Service instance identifiers to match.
owners	List<Name>	Lock owner names to match.
expiresBefore	DateTime	Upper bound of the expiry time.
expiresAfter	DateTime	Lower bound of the expiry time.

4.4 struct OrchestrationLockRequest

Exclusive reservation of a service instance.

Field	Type	Description
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
owner	Name	System for which the instance is locked.
expiresAt	DateTime	UTC expiry of the lock.

4.5 struct OrchestrationLockResponse

A lock on a service instance.

Field	Type	Description
id	Number	Identifier of the lock.
orchestrationJobId	Identifier	Orchestration job that created the lock.
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
owner	Name	Name of the system owning the lock.
expiresAt	DateTime	UTC expiry of the lock.
temporary	Boolean	True for a lock held only during an orchestration.

4.6 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.7 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceorchestration-dynamic/documentation/v5.3.0/DynamicServiceOrchestration-v5.3.0-SysD.pdf.

[2] ——, “serviceOrchestrationLockManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationLockManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationLockManagement MQTT/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationLockManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	